

MATLAB Solution to 2D Time Dependent Schrödinger equation

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1 Solving the 1D Time-Dependent Schrödinger equation

1.1 Theory

The continuum, non-dimensionalized time-dependent one-dimensional Schrödinger equation is:

$$i\psi(x, t)_t = -\psi_{xx} + V(x, t)\psi \quad (1)$$

Where $\psi(x, t)$ is a complex number, which can be represented as a sum of its real and imaginary parts $\psi = \psi_{Re} + i\psi_{Im}$. Equation 1 is to be solved on the domain

$$0 \leq x \leq 1, \quad 0 \leq t \leq t_{max}$$

and subject to the initial and boundary conditions

$$\psi(x, 0) = \psi_0(x), \quad (2)$$

$$\psi(0, t) = \psi(1, t) = 0. \quad (3)$$

A family of exact solution to (1) exists:

$$\psi(x, t) = e^{-im^2\pi^2t} \sin m\pi x \quad (4)$$

where m is a positive integer. This is the exact solution to the 1D time-dependent Schrödinger equation (eq. 1) for initial data type

$$\psi(x, 0) = \sin m\pi x \quad (5)$$

or, as termed in my implementation, `idtype == 0`.

To discretize the continuum domain, we construct a temporal and spatial mesh according to the following: the discretization level l , the ratio of temporal to spatial mesh spacings = $\frac{\Delta t}{\Delta x}$:

$$n_x = 2^l + 1 \quad (6)$$

$$\Delta x = 2^{-l} \quad (7)$$

$$\Delta t = \Delta x \quad (8)$$

$$n_t = \text{round}(t_{max}/\Delta t) + 1 \quad (9)$$

where rounding is used because n_t must be an integer. We can then write the FDA for equations (1)-(3) using the Crank-Nicolson scheme:

$$i \frac{\psi_j^{n+1} - \psi_j^n}{\Delta t} = -\frac{1}{2} \left(\frac{\psi_{j+1}^{n+1} - 2\psi_j^{n+1} + \psi_{j-1}^{n+1}}{\Delta x^2} + \frac{\psi_j^n - 2\psi_j^n + \psi_{j-1}^n}{\Delta x^2} \right) + \frac{1}{2} V_j^{n+\frac{1}{2}} (\psi_j^{n+1} + \psi_j^n) \quad (10)$$
$$j = 2, 3, \dots, n_x - 1, \quad n = 1, 2, \dots, n_t - 1$$

$$\psi_1^{n+1} = \psi_{n_x}^{n+1} = 0, \quad n = 1, 2, \dots, n_t - 1 \quad (11)$$

$$\psi_j^1 = \psi_0(x_j), \quad j = 1, 2, \dots, n_x. \quad (12)$$

We will be using these equations in our implementation of `sch_1d_cn`.

1.2 Method

I implemented my solution of (10)-(12) in a function called `sch_1d_cn`, as instructed. To begin, the function defines the discrete, uniform meshes for x and t according to the `level`, `tmax`, and `lambda` with mesh spacings Δt , Δx and number of temporal/spatial steps n_t , n_x as defined by equations (6)-(9). The x and t arrays are constructed using `linspace`. Then, the initial data type is set according to the given parameters `idtype` and `idpar`. If `idpar == 0`, the initial data type (i.e. the value of $\psi(x, t = 0)$) are calculated according to equation (5), where m is the first element of the array `idpar`. If `idpar == 1`, then the initial data type is a boosted Gaussian:

$$\psi(x, t = 0) = e^{ipx} e^{-((x-x_0)/\delta)^2} \quad (13)$$

where x_0 , δ , and p are given by, respectively, the first, second, and third elements of `idpar`.

The next step in my implementation is the determination of the potential term. If `vtype` is given to be 0, then there is no potential: the potential term is an array of zeros of size $1 \times n_x$. If `vtype` is given to be 1, then the potential is as defined for a rectangular barrier or well:

$$V(x) = \begin{cases} 0 & \text{for } x < x_{min} \\ V_c & \text{for } x_{min} \leq x \leq x_{max} \\ 0 & \text{for } x > x_{max} \end{cases} \quad (14)$$

where V_c is a constant, $V_c < 0$ for a well and $V_c > 0$ for a barrier.

The final step in my implementation was to step up the tridiagonal system and to solve equations (10)-(12) using the Crank-Nicolson scheme. To do this, I had to rearrange equation (10) such that all ψ^{n+1} terms were on the RHS and all ψ^n terms were on the LHS and the equation takes the form:

$$C_j^+ \psi_{j+1}^{n+1} + C_j^0 \psi_j^{n+1} + C_j^- \psi_{j-1}^{n+1} = f_j. \quad (15)$$

Doing this gives the following:

$$C_j^+ = \frac{1}{2\Delta x^2} \quad (16)$$

$$C_j^0 = \frac{i}{\Delta t} - \frac{1}{\Delta x^2} - \frac{1}{2} V_2^{n+1/2} \quad (17)$$

$$C_j^- = \frac{1}{2\Delta x^2} \quad (18)$$

$$f_j = i \frac{\psi_j^n}{\Delta t} - \frac{1}{2} \left(\frac{\psi_j^n - 2\psi_j^n + \psi_{j-1}^n}{\Delta x^2} \right) + \frac{1}{2} V_2^{n+1/2} \psi_j^n \quad (19)$$

$$j = 2, 3, \dots, n_x - 1, \quad n = 1, 2, \dots, n_t - 1$$

Note that this only defines the interior grid points of the mesh, and the boundary points are defined as stated by the boundary conditions (equation (3)). Therefore, $f_{j=0} = f_{j=n_x} = 0$ in my implementation. I construct a diagonal matrix A of size $n_x \times n_x$ of the form:

$$A = \begin{pmatrix} 1 & 0 & 0 & \dots & \dots & \dots & 0 \\ * & * & * & 0 & \dots & \dots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 & \dots & \vdots \\ \vdots & 0 & C_j^- & C_j^0 & C_j^+ & 0 & \vdots \\ \vdots & \dots & 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \dots & \dots & 0 & * & * & * \\ 0 & \dots & \dots & \dots & 0 & 0 & 1 \end{pmatrix} \quad (20)$$

where the non-zero entries (indicated by asterisk and diagonal dots) on the lower-diagonal, the diagonal, and the upper-diagonal are defined, respectively by C_j^- , C_j^0 , and C_j^+ (not including the boundaries). The matrix was constructed with `spdiags` to speed up the program.

I then iterate over time t_n where $n = 1, 2, \dots, n_t - 1$. At each iteration, I calculate f_j according to equation (19), setting $f_{j=0} = f_{j=n_x} = 0$, then I calculate $\psi_j^{n+1} = A \setminus f_j$ where $\setminus$ is the left division operator. If the potential was time-dependent, I would compute the matrix A at every time step using the potential term at that time step. And the potential term in f_j would also change at every time step.

In addition to returning the vector of x coordinates, the vector of t coordinates, the array of potential values v , and the array of computed ψ values, `sch_1d_cn` also returns an array of the real components of the computed psi values (determined using `real(...)`), an array of the imaginary components of the computed psi values (determined using `imag(...)`), a useful diagnostic: an array of the square-rooted probability density `psimod` which is defined and computed as:

$$\text{psimod} = \sqrt{\rho} = |\psi| = \sqrt{\psi\psi^*} \quad (21)$$

where ψ^* is the complex conjugate of ψ , and another useful diagnostic: the probability `prob` which is defined as the “running integral” of the probability density ρ and approximated to $O(h^2)$ using the trapezoidal formula:

$$P(x_j, t^n) = \int_{x_0}^{x_j} \psi(\tilde{x}, t^n) \psi^*(\tilde{x}, t^n) d\tilde{x} \approx \frac{1}{2} \sum_{i=1}^{j-1} (P_i + P_{i+1})(x_{i+1} - x_i). \quad (22)$$

1.3 Results and Analysis

1.3.1 Convergence Tests

If we define a level l solution computed using my implementation of `sch_1d_cn` as ψ^l . And we denote the difference

$$d\psi^l = \psi^{l+1} - \psi^l$$

. where ψ^{l+1} has been 2:1 coarsened in both space and time dimensions such that it has the same size as ψ^l . To test convergence we will be computing the $l - 2$ norm of this quantity:

$$\|d\psi^l\|_2 = \sqrt{\frac{\sum_{j=1}^{N-1} |d\psi^{lj}|^2}{N}} \quad (23)$$

where N is the product of the dimensions of array $d\psi^l$ (i.e. it's length) and where $|d\psi^{lj}|$ is the modulus of $d\psi^{lj}$. This quantity is a function of discrete time t^n with a length of n_t . Since our FDA is $O(h^2)$ accurate, where $(\Delta x = h$ and $\Delta t = \lambda h)$ so the expansion of ψ^l must take the form

$$\psi^l(x, t) = \psi(x, t) + h_l^2 e_2(x, t) + O(h_l^4)$$

so the plots of the scaled $\|d\psi^l\|_2$ and our solution convergences, we should see coincidence of the curves, with better agreement as level increases. For both of our convergence tests, we will be using levels 6-9, with $\|d\psi^{l7}\|_2$ being scaled by 4, and $\|d\psi^8\|_2$ being scaled by 16. I implement the test in `prob1convergence` for a solution to the 1D time-dependent Schrödinger approximated by `sch_1d_cn` with the following input parameters:

- `idtype = 0`
- `vtype = 0`
- `idpar = [3]`
- `tmax = 0.25`
- `lambda = 0.1`

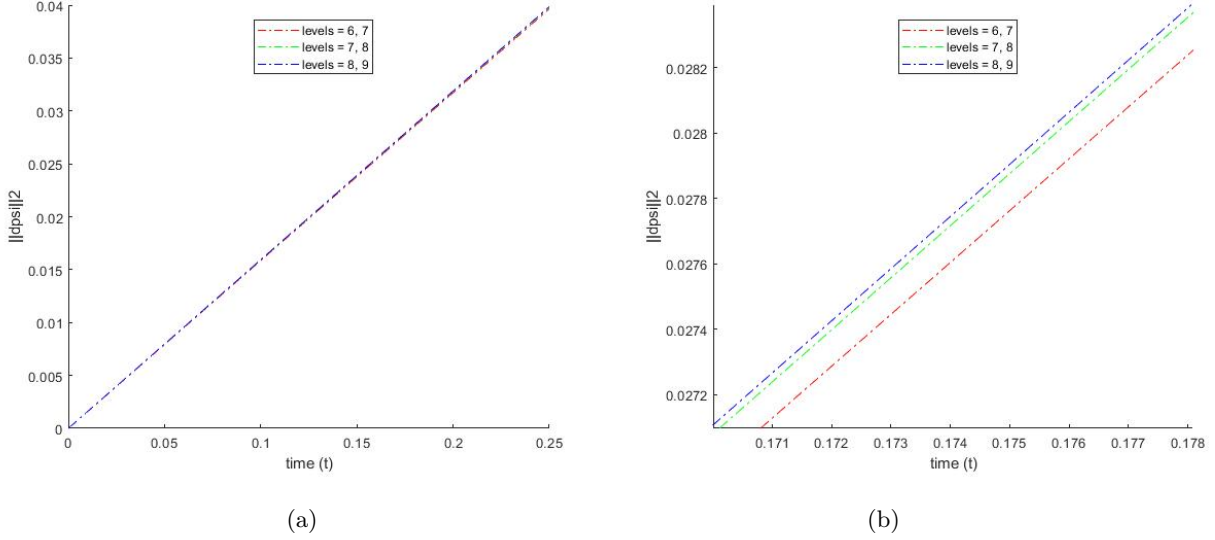


Figure 1: Convergence plot for $\|d\psi^l\|_2$ of a solution to the 1D time-dependent Schrödinger for the following input parameters: `idtype=0`, $m = 3$, no potential, $t_{max} = 0.25$, $\lambda = 0.1$ for levels 6-9. $\|d\psi^6\|_2$ (red) was not scaled, $\|d\psi^7\|_2$ (green) was scaled by a factor of 4, $\|d\psi^8\|_2$ (blue) was scaled by a factor of 16. $\psi^{l=7}$ was down-sampled by 2, $\psi^{l=8}$ was down-sampled by 4, and $\psi^{l=9}$ was down-sampled by 8. (b) shows a zoomed version of (a).

and the resulting plot is given below in figure 1.

Since for `idtype = 0` we know the exact solution, we compute it at each level using equation (4), and then compute the error norm:

$$\|E(\psi^l)\|_2(t^n) = \|\psi_{exact} - \psi^l\|_2(t^n) \quad (24)$$

I included the plot of this for the above solution as figure 2. Note that, as with the difference norm, the error norm is scaled before it is plotted. $\|E(\psi^7)\|_2$ is scaled by 4, $\|E(\psi^8)\|_2$ is scaled by 16, and $\|E(\psi^9)\|_2$ is scaled by 64.

Both of these graphs (difference norm and error norm) show near coincidence of the curves, with better agreement at higher levels, which is, as stated previously, what we would expect from a converging solution.

Additionally, in figures 3-4, I include a snapshot of both the real component of the solution and the imaginary component of the solution at $t = 6.25 \times 10^{-3}$ plotted with the exact solution. Note that, as level increases the approximation approaches the exact solution.

I then repeated the same 4-level convergence test for a solution with the parameters:

- `idtype = 1`
- `vtype = 0`
- `idpar = [0.50 0.075 0.0]`
- `tmax = 0.01`
- `lambda = 0.01`

`idtype = 1` has no exact solution, so I was unable to conduct the error norm test. However the results of the $\|d\psi^l\|_2$ (see figure 4) were as I expected for a converging solution, the curves were coincident, with better agreement as level increased.

I also plotted a snapshot of ψ_{Re} (fig. 6) and ψ_{Im} (fig. 7) at $t = 7.5 \times 10^{-3}$. The curves coincided nicely, with better agreement as level increased.

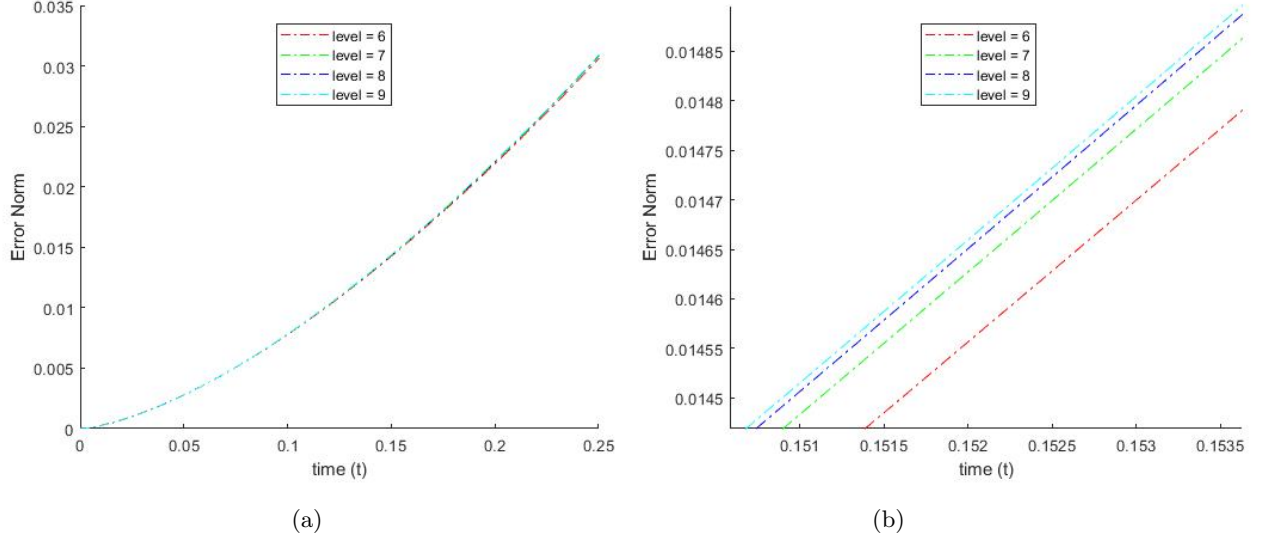


Figure 2: Plot of Error norm $\|E(\psi^l)\|_2$ as a function of time of a solution to the time-dependent Schrödinger for the following input parameters: `idtype=0`, $m = 3$, no potential, $t_{max} = 0.25$, $\lambda = 0.1$ for levels 6-9. $\|E(\psi^7)\|_2$ is scaled by 4, $\|E(\psi^8)\|_2$ is scaled by 16, and $\|E(\psi^9)\|_2$ is scaled by 64. (b) shows a zoomed version of (a).

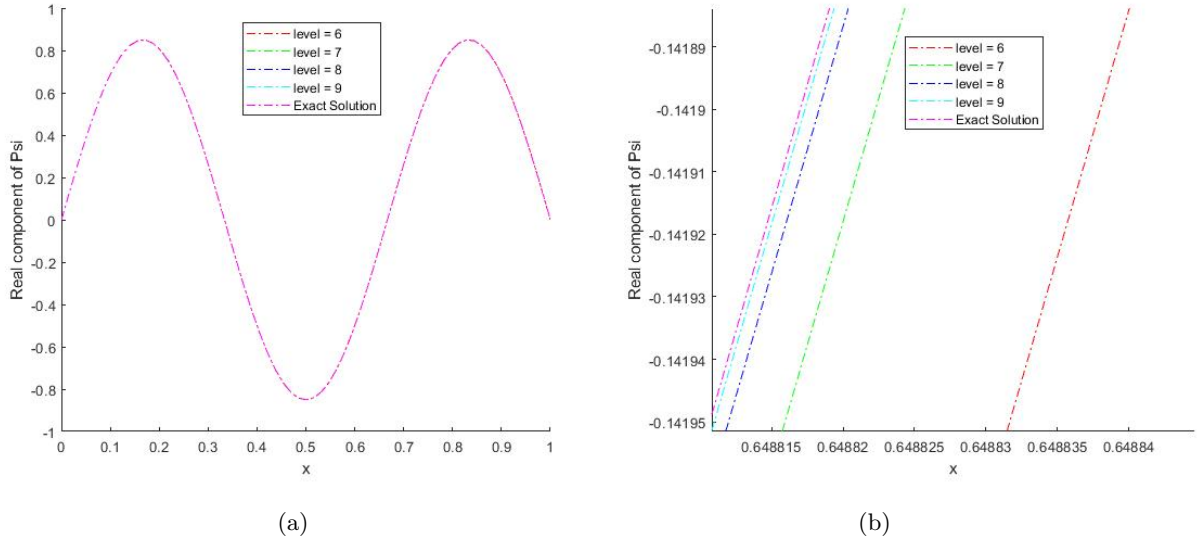


Figure 3: Plot of $\psi_{Re}(x, t = 6.25 \times 10^{-3})$ for 4 different level approximations for `idtype == 0`, `vtype == 0`, $\lambda = 0.1$, $t_{max} = 0.25$, and $m = 3$. The real component of the exact solution at this time (see equ. 2) was also plotted. Level 7 was down-sampled by 2, level 8 was down-sampled by 4, and level 9 was down-sampled by 8. (b) shows a zoomed version of (a).

1.3.2 Experiments

I implemented the numerical experiments in the script `problexperiments`. These experiments were conducted by looking at `prob`, defined earlier by equation (22). For every x I took the temporal average $\bar{P}_j$:

$$\bar{P}_j = \frac{\sum_{n=1}^{n_t} P_j^n}{n_t}$$

then normalized by dividing the whole array by the last value:

$$\bar{P}_j := \bar{P}_j / \bar{P}_{n_x}.$$

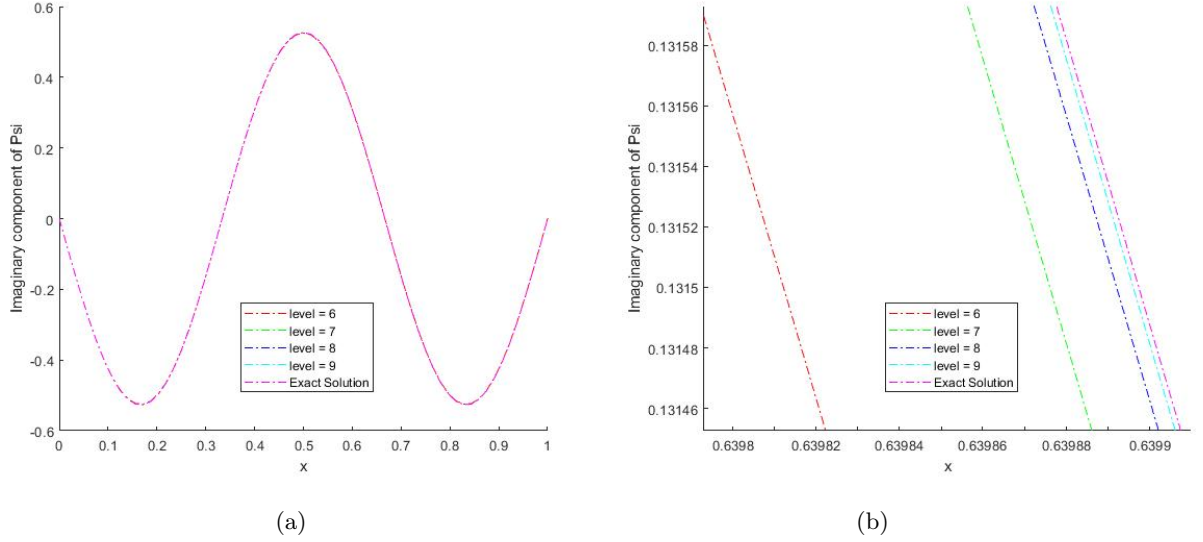


Figure 4: Plot of $\psi_{Im}(x, t = 6.25 \times 10^{-3})$ for 4 different level approximations for `idtype == 0`, `vtype == 0`, $\lambda = 0.1$, $t_{max} = 0.25$, and $m = 3$. The imaginary component of the exact solution at this time (see equ. 2) was also plotted. Level 7 was down-sampled by 2, level 8 was down-sampled by 4, and level 9 was down-sampled by 8. (b) shows a zoomed version of (a).

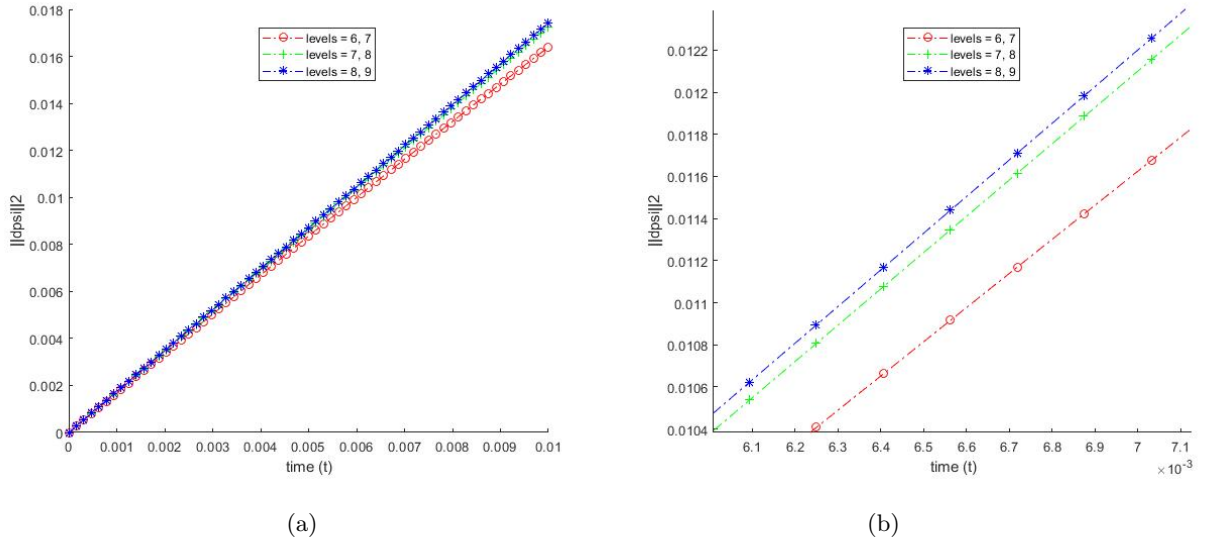


Figure 5: Convergence plot for $\|d\psi^l\|_2$ of a solution to the 1D time-dependent Schrödinger for the following input parameters: `idtype=1` (Gaussian), $x_0 = 0.50$, $\delta = 0.075$, $p = 0.0$, no potential, $t_{max} = 0.01$, $\lambda = 0.01$ for levels 6-9.. $\|d\psi^6\|_2$ (red) was not scaled, $\|d\psi^7\|_2$ (green) was scaled by a factor of 4, $\|d\psi^8\|_2$ (blue) was scaled by a factor of 16. $\psi^{l=7}$ was down-sampled by 2, $\psi^{l=8}$ was down-sampled by 4, and $\psi^{l=9}$ was down-sampled by 8. (b) shows a zoomed version of (a).

We can then interpret

$$\bar{P}(x_2) - \bar{P}(x_1)$$

as the fraction of time the quantum particle spends in the interval $x_1 \leq x \leq x_2$. For a free particle, at sufficiently large times we expect that

$$\bar{P}(x_2) - \bar{P}(x_1) \rightarrow x_2 - x_1$$

as a free particle is just as likely to be at any x . For any non-zero potential, the excess fractional probability

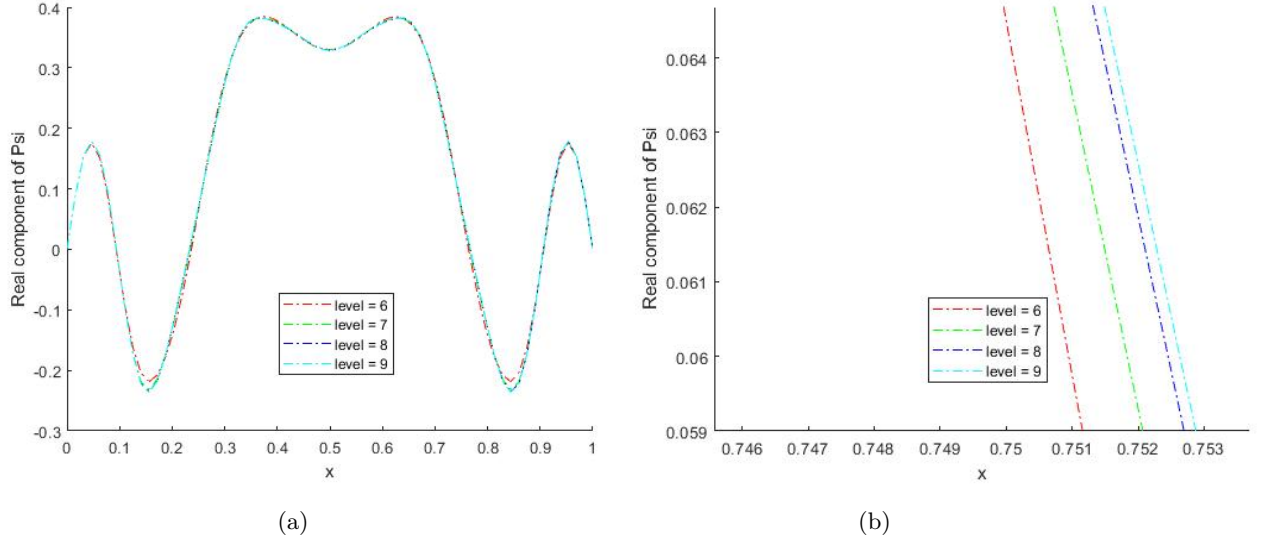


Figure 6: Plot of $\psi_{Re}(x, t = 7.5 \times 10^{-3})$ for 4 different level approximations for `idtype == 1`, `vtype == 0`, $\lambda = 0.01$, $t_{max} = 0.01$, $x_0 = 0.50$, $\delta = 0.075$, and $p = 0.0$. Level 7 was down-sampled by 2, level 8 was down-sampled by 4, and level 9 was down-sampled by 8. (b) shows a zoomed version of (a).

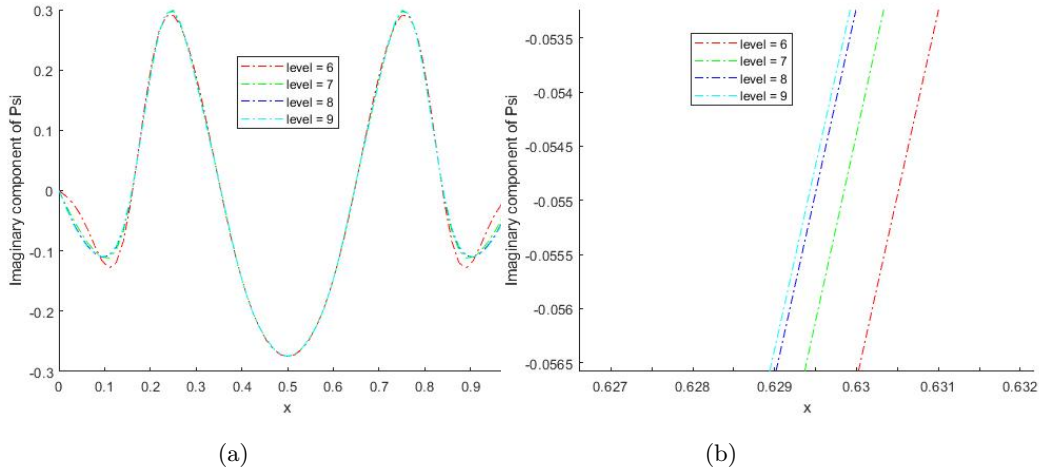


Figure 7: Plot of $\psi_{Im}(x, t = 7.5 \times 10^{-3})$ for 4 different level approximations for `idtype == 1`, `vtype == 0`, $\lambda = 0.01$, $t_{max} = 0.01$, $x_0 = 0.50$, $\delta = 0.075$, and $p = 0.0$. Level 7 was down-sampled by 2, level 8 was down-sampled by 4, and level 9 was down-sampled by 8. (b) shows a zoomed version of (a).

that the particle spends in a given spatial interval is:

$$\bar{F}_e(x_1, x_2) = \frac{\bar{P}(x_2) - \bar{P}_1}{x_2 - x_1}.$$

If this quantity is less than one (which will be the case for these experiments) this indicates that the particle is spending less time in the given interval than a free particle would. For our purposes, we will be plotting $\ln \bar{F}_e(x_1, x_2)$ as a function of $\ln V_0$.

For the first experiment, I consider a barrier potential. The ‘height’ of this barrier V_0 is the control parameter for this experiment. I used 251 uniformly spaced values of $\ln V_0$ in the range -2 to 5. The parameters for the experiment were as follows:

- `idtype = 1`
- `vtype = 1`

- `idpar = [0.40 0.075 20.0]`
- `idpar = [0.6 0.8 VARIABLE > 0]`
- `tmax = 0.10`
- `lambda = 0.01`
- `level = 9`
- `x1 = 0.8`
- `x2 = 1.0`

The results are plotted in figure 8.

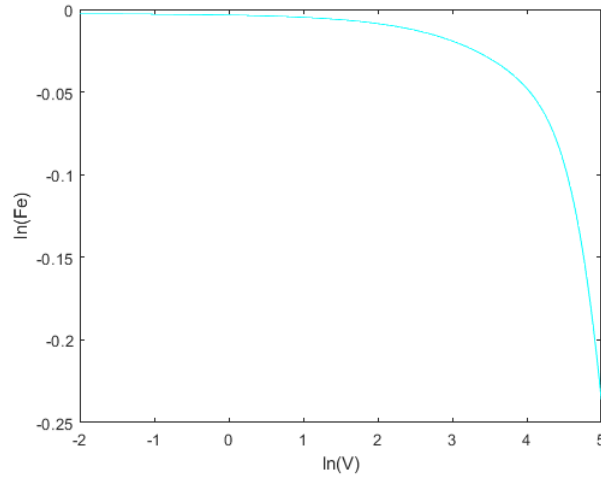


Figure 8: Plot of $\ln \bar{F}_e(x_1, x_2)$ as a function of $\ln V_0$ for a barrier potential.

From figure 8, we can gather that as potential increases, the quantum particle spends a lot less time in the barrier. It seems to approach $-\infty$, which means that, for an infinitely large potential, the particle would spend no time within the bounds of the barrier or beyond the barrier (since we are considering an interval $x_1 = 0.8$ to $x_1 = 1.0$) at all.

For the second experiment, I consider a well potential. The ‘depth’ of this well V_0 is the control parameter for this experiment. I used 251 uniformly spaced values of $\ln |V_0|$ in the range 2 to 10. Note that V_0 is strictly negative. The parameters for the experiment were as follows:

- `idtype = 1`
- `vtype = 1`
- `idpar = [0.40 0.075 20.0]`
- `idpar = [0.6 0.8 VARIABLE < 0]`
- `tmax = 0.10`
- `lambda = 0.01`
- `level = 9`
- `x1 = 0.6`
- `x2 = 0.8`

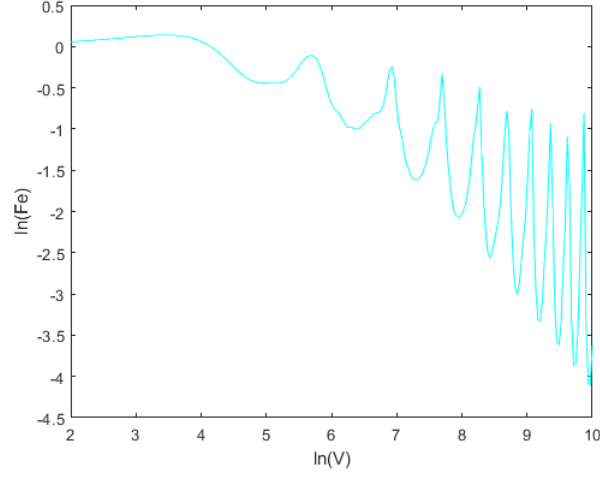


Figure 9: Plot of $\ln \bar{F}_e(x_1, x_2)$ as a function of $\ln V_0$ for a well potential.

The results are plotted in figure 9.

From figure 9, we can gather that as potential increases, we get oscillations. I interpret this as the particle flowing in and out of the well, eventually becoming trapped in the well. For an infinite potential, the particle would become fully trapped in the well.

The implementation of these experiments can be found in the script `prob1experiments`.

2 Solving the 2D Time Dependent Schrödinger equation

2.1 Theory

The continuum, non-dimensionalized time-dependent 2D Schrödinger equation multiplied through by $-i$ is:

$$\psi(x, y, t)_t = i(\psi_{xx} + \psi_{yy}) - iV(x, y)\psi \quad (25)$$

to be solved on the domain:

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1, \quad 0 \leq t \leq t_{max}$$

and subject to the initial and boundary conditions

$$\psi(x, y, 0) = \psi_0(x, y), \quad (26)$$

$$\psi(0, y, t) = \psi(1, y, t) = \psi(x, 0, t) = \psi(x, 1, t) = 0. \quad (27)$$

A family of exact solution to (25) exists:

$$\psi(x, t) = e^{-i(m_x^2 + m_y^2)\pi^2 t} \sin(m_x \pi x) \sin(m_y \pi y) \quad (28)$$

where m_x and m_y are positive integers. This is the exact solution to the 1D time-dependent Schrödinger equation (eq. 1) for initial data type

$$\psi(x, y, 0) = \sin(m_x \pi x) \sin(m_y \pi y) \quad (29)$$

or, as termed in my implementation, `idtype == 0`.

To discretize the continuum domain, we construct a temporal and spatial meshes according to the following: the discretization level l , the ratio of temporal to spatial mesh spacings $= \frac{\Delta t}{\Delta x} = \frac{\Delta t}{\Delta y}$:

$$n_x = n_y = 2^l + 1 \quad (30)$$

$$\Delta x = \Delta y = 2^{-l} \quad (31)$$

$$\Delta t = \Delta x \quad (32)$$

$$n_t = \text{round}(t_{max}/\Delta t) + 1 \quad (33)$$

where rounding is used because n_t must be an integer and where n_x and n_y are equal. We can then write the FDA for equations (25)-(27) using the alternating direction implicit (ADI) method:

$$\left(1 - i\frac{\Delta t}{2}\partial_{xx}^h\right)\psi_{i,j}^{n+\frac{1}{2}} = \left(1 + i\frac{\Delta t}{2}\partial_{xx}^h\right)\left(1 + i\frac{\Delta t}{2}\partial_{yy}^h - i\frac{\Delta t}{2}V_{i,j}\right)\psi_{i,j}^n, \quad (34)$$

$$i = 2, 3, \dots, n_x - 1, \quad j = 2, 3, \dots, n_y - 1, \quad n = 1, 2, \dots, n_t - 1$$

$$\left(1 - i\frac{\Delta t}{2}\partial_{yy}^h + i\frac{\Delta t}{2}V_{i,j}\right)\psi_{i,j}^{n+1} = \psi_{i,j}^{n+\frac{1}{2}},$$

$$i = 2, 3, \dots, n_x - 1, \quad j = 2, 3, \dots, n_y - 1, \quad n = 1, 2, \dots, n_t - 1 \quad (35)$$

where the difference operators ∂_{xx}^h and ∂_{yy}^h are defined by:

$$\partial_{xx}^h u_{i,j}^n \equiv \frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} \quad (36)$$

$$\partial_{yy}^h u_{i,j}^n \equiv \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} \quad (37)$$

with initial conditions:

$$\psi_{i,j}^1 = \psi_0(x_i, y_j) \quad (38)$$

and boundary conditions:

$$\psi_{1,j}^n = \psi_{n_x,j}^n = \psi_{i,1}^n = \psi_{i,n_y}^n = 0. \quad (39)$$

2.2 Method

I implemented my solution to (34)-(39) in the function `sch_2d_adi`. I began with constructing the spatial and temporal meshes according to equations (30)-(33), then I initialized my output arrays. Next, I initialized the solution `psi` with the initial data according to the given parameters `idtype` and `idpar`. If `idpar == 0`, the initial data type (i.e. the value of $\psi(x, y, t = 0)$) are calculated according to equation (29), where m_x and m_y are, respectively, the first and second elements of the array `idpar`. If `idpar == 1`, then the initial data type is a boosted Gaussian:

$$\psi(x, y, t = 0) = e^{ip_x x} e^{ip_y y} e^{-((x-x_0)^2/\delta_x^2 + (y-y_0)^2/\delta_y^2)} \quad (40)$$

where x_0 , y_0 , δ_x , δ_y , p_x , and p_y are given by, respectively, the first, second, third, fourth, fifth, and sixth elements of `idpar`.

The next step in my implementation is the determination of the potential term. If `vtype` is given to be 0, then there is no potential: the potential term is an array of zeros of size $1 \times n_x \times n_y$. If `vtype` is given to be 1, then the potential is as defined for a rectangular barrier or well:

$$V(x, y) = \begin{cases} V_c & \text{for } (x_{min} \leq x \leq x_{max}) \text{ and } (y_{min} \leq y \leq y_{max}) \\ 0 & \text{otherwise} \end{cases} \quad (41)$$

where V_c is a constant, $V_c < 0$ for a well and $V_c > 0$ for a barrier. x_{min} , x_{max} , y_{min} , y_{max} , and V_c are given by the array `vtype`.

The final step in my implementation was to step up the tridiagonal systems and to solve equations (34)-(39) using the ADI method. To do this, I had to construct 2 different tridiagonal systems of the form outlined in equation (15). The first was based on equation (34). Rearranging (34) into the form (15) gives the following:

$$C_i^+ = -i \frac{\Delta t}{2\Delta x^2} \quad (42)$$

$$C_i^0 = 1 + i \frac{\Delta t}{\Delta x^2} \quad (43)$$

$$C_i^- = -i \frac{\Delta t}{2\Delta x^2} \quad (44)$$

$$f_{half} = \left(1 + i \frac{\Delta t}{2} \partial_{xx}^h\right) \left(1 + i \frac{\Delta t}{2} \partial_{yy}^h - i \frac{\Delta t}{2} V_{i,j}\right) \quad (45)$$

$$i = 2, 3, \dots, n_x - 1, \quad n = 1, 2, \dots, n_t - 1$$

where ∂_{xx}^h and ∂_{yy}^h are as defined in equations (36)-(37) and $f_{i=0} = f_{i=n_x} = 0$ due to the boundary conditions. Using this, I construct a diagonal matrix A_{half} of size $n_x \times n_x$ of the form outlined in equation (20). For f_i , it is imperative to note that ∂_{xx}^h acts on the result of the second bracket of the LHS of equation (45).

I then calculated the second tridiagonal system in much the same method but using equation (35). This gives the result:

$$C_j^+ = -i \frac{\Delta t}{2\Delta y^2} \quad (46)$$

$$C_j^0 = 1 + i \frac{\Delta t}{\Delta y^2} + i \frac{\Delta t}{2} V_{i,j} \quad (47)$$

$$C_j^- = -i \frac{\Delta t}{2\Delta y^2} \quad (48)$$

$$f = 1 \quad (49)$$

$$j = 2, 3, \dots, n_y - 1, \quad n = 1, 2, \dots, n_t - 1$$

Since A has a dependence on $V_{i,j}$, I changed C_j^0 at every iteration over i .

I iterate over the discrete time mesh t_n for $n = 1, 2, \dots, n_t - 1$ and at every iteration there are 2 stages to the calculation. The first solves the first tridiagonal system $\psi_{i,j}^{n+\frac{1}{2}} = A_{half} \backslash f_{half}$ for each $j = 2, 3, \dots, n_y - 1$ for $\psi_{i,j}^{n+\frac{1}{2}}, i = 1, 2, \dots, n_x$. The second stage takes this result for each $i = 2, 3, \dots, n_x - 1$, sets it equal to f , calculates A according to (), and solves $\psi_{i,j}^n = A \backslash f$ for $\psi_{i,j}^n, j = 1, 2, \dots, n_y$. This is our final result.

2.3 Results and Analysis

2.3.1 Convergence Test: Free Particle

In the script `prob2convergenctest.m` I performed a 4-level (levels 6-9) convergence test of `sch_2d_adi` for a free particle with parameters:

- `idtype = 0`
- `vtype = 0`
- `idpar = [2 3]`
- `tmax = 0.05`
- `lambda = 0.05`
- `lmin = 6`
- `lmin = 9`

I followed the same method outlined in 1.3.1, calculating the difference norm $\|d\psi^l\|_2$ according to equation (23) and the error norm $\|E(\psi^l)\|_2$ according to equation (24) with the exact solution defined by equation (28). Again, $\|d\psi^7\|_2$ was scaled by 4, and $\|d\psi^8\|_2$ was scaled by 16, $\|E(\psi^7)\|_2$ was scaled by 4, $\|E(\psi^8)\|_2$ was scaled by 16, and $\|E(\psi^9)\|_2$ was scaled by 64. The results of the tests are shown in figures 10-11.

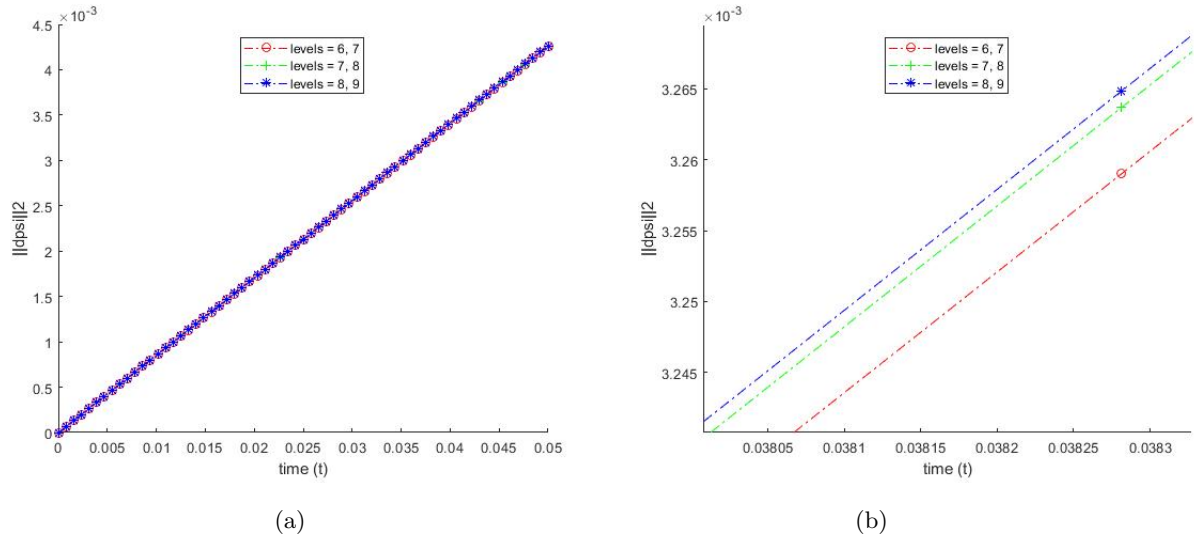


Figure 10: Convergence plot for $\|d\psi^l\|_2$ of a solution to the 2D time-dependent Schrödinger for the following input parameters: `idtype=0`, $m_x = 2$, $m_y = 3$, no potential, $t_{max} = 0.05$, $\lambda = 0.05$ for levels 6-9. $\|d\psi^6\|_2$ (red) was not scaled, $\|d\psi^7\|_2$ (green) was scaled by a factor of 4, $\|d\psi^8\|_2$ (blue) was scaled by a factor of 16. $\psi^{l=7}$ was down-sampled by 2, $\psi^{l=8}$ was down-sampled by 4, and $\psi^{l=9}$ was down-sampled by 8. (b) shows a zoomed version of (a).

Both of these tests demonstrate convergence. The curves are all coincident with better agreement as level is increased.

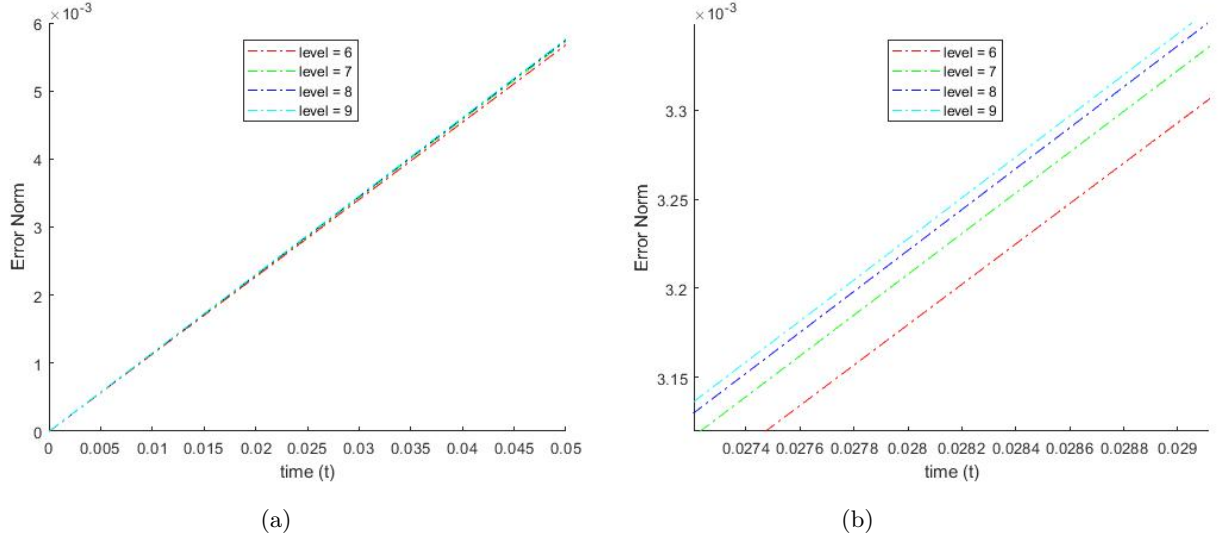


Figure 11: Plot of Error norm $\|E(\psi^l)\|_2$ as a function of time of a solution to the 2D time-dependent Schrödinger for the following input parameters: `idtype=0`, $m_x = 2$, $m_y = 3$, no potential, $t_{max} = 0.05$, $\lambda = 0.05$ for levels 6-9. $\|E(\psi^7)\|_2$ is scaled by 4, $\|E(\psi^8)\|_2$ is scaled by 16, and $\|E(\psi^9)\|_2$ is scaled by 64. (b) shows a zoomed version of (a).

2.3.2 Experiment: Rectangular Barrier

For the rectangular barrier I used the following parameters:

- `idtype = 1`
- `vtype = 1`
- `idpar = [0.8 0.5 0.1 0.1 0.0 0.0]`
- `vpar = [0.1 0.3 0.0 1.0 1000]`
- `tmax = 0.05`
- `lambda = 0.01`
- `level = 9`

I plotted the potential (figure 12) to ensure correct placement, then I made the video at level 9. As you can see in figure 13 (b), the barrier effectively blocked the wave packet from passing it. I didn't see this so much with potentials on the order of unity, and I was a little confused for a while as to the reason why. After giving it some thought though, I considered my results from my problem 1 experiments. The probability of the quantum particle being in the barrier is still fairly high for unity values of potential, to get a very low probability (to stop the particle entirely, I would have to increase potential by quite a bit, which I did. The video for this experiment is labelled `sch_barrier.avi` and the script for this experiment is `prob2experiment_barrier.m`.

2.3.3 Experiment: Rectangular Well

For the rectangular well I used the following parameters:

- `idtype = 1`
- `vtype = 1`

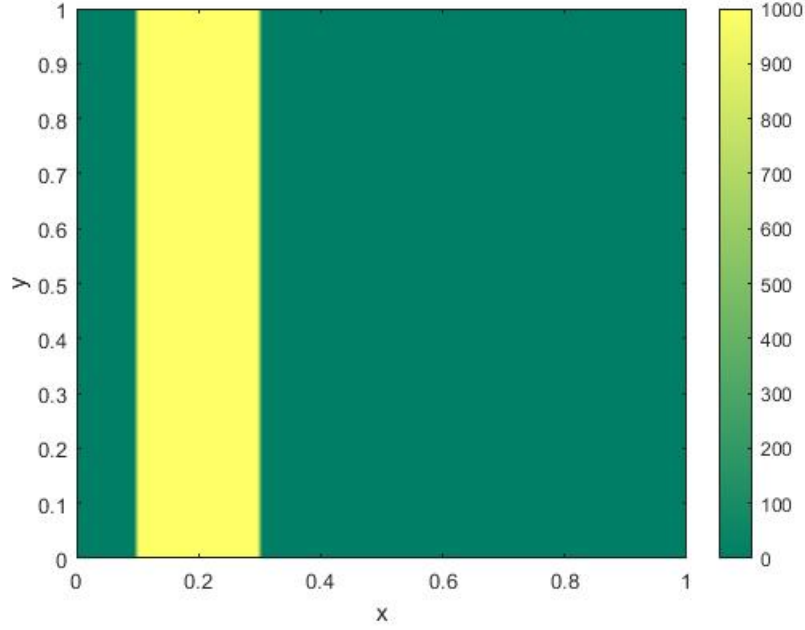


Figure 12: Graph of the time-independent potential of the rectangular barrier experiment.

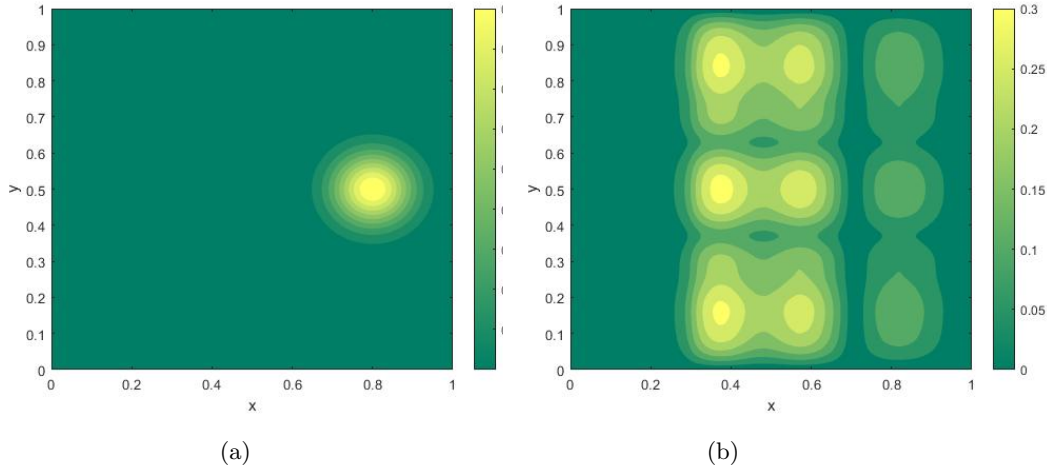


Figure 13: Screenshots of the initial (a) and final (b) states of the rectangular barrier experiment from the video sch_barrier.avi

- `idpar = [0.8 0.5 0.1 0.1 0.0 0.0]`
- `vpar = [0.1 0.3 0.0 1.0 -1000]`
- `tmax = 0.05`
- `lambda = 0.01`
- `level = 9`

I structured the well experiment in the same way as my barrier experiment, but with a negative potential (see fig. 14). I decided to not give the particle any boost (p_x, p_y) because I wanted to see if a stationary particle would pool in the well. And indeed, this is what I witnessed. Even though the quantum particle began with no initial momentum, it slowly moved towards the well, eventually being partially trapped. It then began to resonate within the well, much in the way a free particle outside a well would. Figure 15 (b)

shows the final snapshot of the video, where you can see 9 distinct nodes. The video for this experiment is labelled `sch_well.avi` and the script for this experiment is `prob2experiment_well.m`.

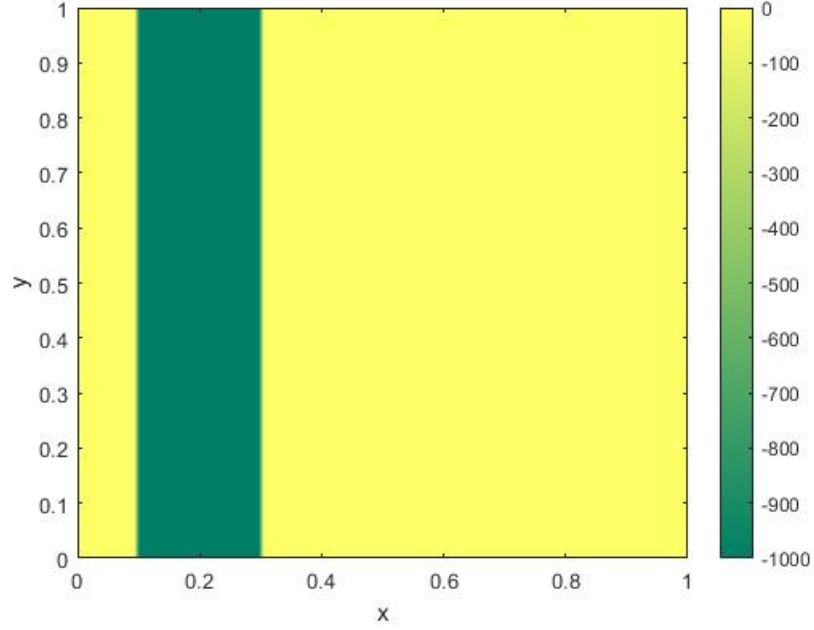


Figure 14: Graph of the time-independent potential of the rectangular well experiment.

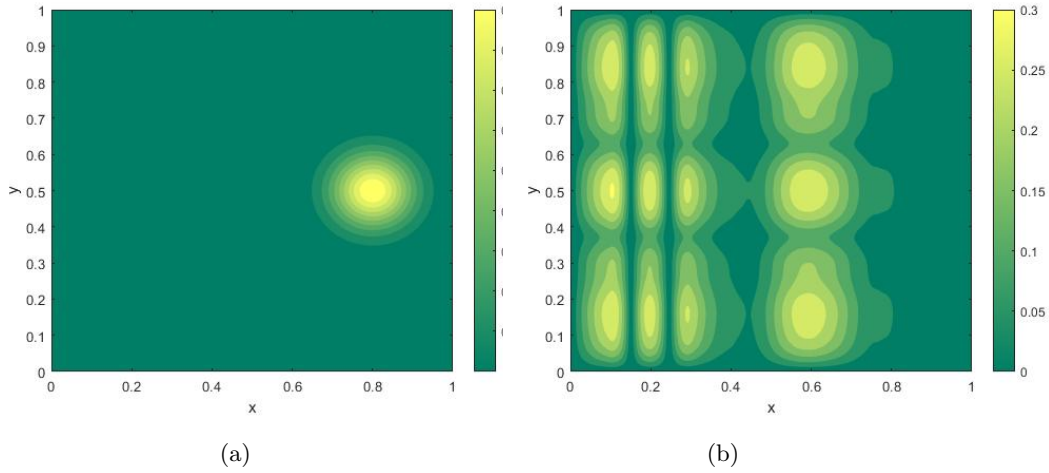


Figure 15: Screenshots of the initial (a) and final (b) states of the rectangular well experiment from the video `sch_well.avi`

2.3.4 Experiment: Double Slit

For the double slit experiment I used the following parameters:

- `idtype = 1`
- `vtype = 2`
- `idpar = [0.5 0.1 0.9 0.05 0 2]`
- `vpar = [0.33 0.37 0.63 0.67 4000]`

- `tmax = 0.03`
- `lambda = 0.01`
- `level = 9`

Since the y -coordinates of the slit were fixed in the implementation of `sch_2d_adi` to $y_{j'}-y_{j'+1}$, I placed my Gaussian quantum particle near the bottom, right under the slit (see figure 16 for the potential, fig 17 (a) for the initial position of the particle). I also gave it a boost in the positive y direction to give it some velocity as it travelled through the slit. It formed the tell-tale double slit diffraction pattern, forming a sort of grating-looking pattern with higher amplitudes towards the middle. This can even be seen in the final snapshot of the video in figure 17 (b). The video for this experiment is labelled `sch_dbslit.avi` and the script for this experiment is `prob2experiment_dbslit.m`.

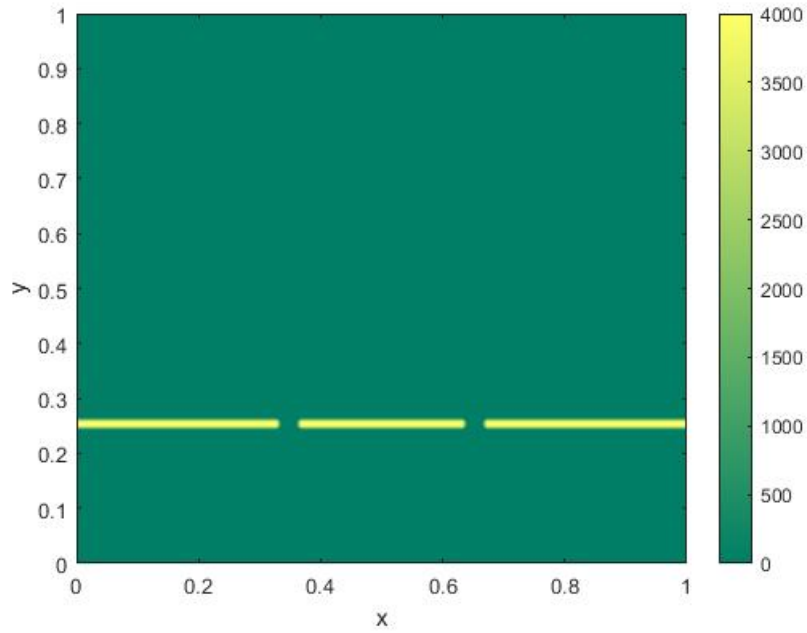


Figure 16: Graph of the time-independent potential of the double slit experiment.

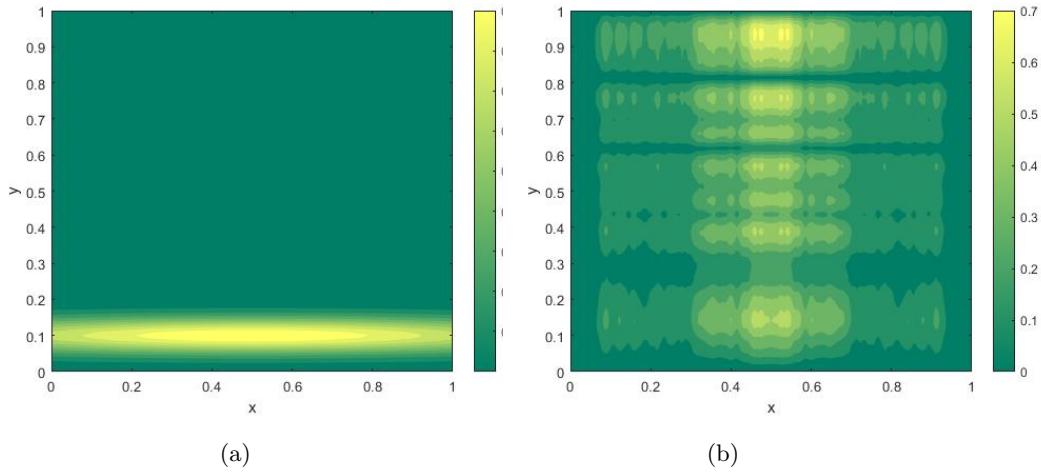


Figure 17: Screenshots of the initial (a) and final (b) states of the double slit experiment from the video `sch_dbslit.avi`